



3 The integral I_n is defined by $I_n = \int_0^1 (1+x^2)^n dx$.

(a) By considering $\frac{d}{dx}(x(1+x^2)^n)$, or otherwise, show that

$$(2n+1)I_n = 2^n + 2nI_{n-1}. \quad [5]$$

[illegible]



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(b) Find the exact value of I_{-2} .

[4]

[illegible]